

Full Test — Answers

Final answers with a one-line reason. For full methods, revisit the concept notes.

Objective

Q1 (b) **64.** $64 = 8^2$.

Q2 (b) **14.** $14^2 = 196$.

Q3 (b) **3.** Units digit 7, and $7^3 = 343$ ends in 3.

Q4 (b) **7.** $7^3 = 343$.

Q5 **14.** Numbers between n^2 and $(n+1)^2 = 2n = 2 \times 7$.

Q6 **False.** $\sqrt{1000} \approx 31.6$; also 1000 has 3 (an odd number of) trailing zeros.

Q7 (c) **100.** $27 = 3^3$, $64 = 4^3$, $125 = 5^3$; 100 is not a cube.

Q8 (a) **Both true, R explains A.** $441 = 21^2$, which directly shows it is a perfect square.

Short answer

Q9 **45.** $2025 = 3^4 \times 5^2 = (3 \times 3)(3 \times 3)(5 \times 5)$, so $\sqrt{} = 3 \times 3 \times 5 = 45$.

Q10 (i) **9** (ii) **1.** $9^3 = 729$ ends in 9; $1^3 = 1$ ends in 1.

Q11 **No.** $2800 = 2^4 \times 5^2 \times 7$; the 7 has no pair, so it is not a perfect square.

Q12 **$64 = 13 + 15 + 17 + 19$.** (The 4th group of consecutive odd numbers = 4^3 .)

Long answer

Q13 **Multiply by 2;** $\sqrt{784} = 28$. $392 = 2^3 \times 7^2$; the extra 2 is unpaired, so $\times 2 \rightarrow 784 = 28^2$.

Q14 **Multiply by 25.** $1080 = 2^3 \times 3^3 \times 5$; the 5 needs two more 5s, so $\times 5^2 = 25 \rightarrow 27000 = 30^3$.

Q15 **900.** $\text{LCM}(4, 9, 10) = 180 = 2^2 \times 3^2 \times 5$; multiply by 5 to pair the 5 $\rightarrow 900 = 30^2$.

Q16 **49 rows.** Rows = $\sqrt{2401} = \sqrt{(7^4)} = 7^2 = 49$.

Total: 28 marks.